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ECG Arrhythmia Classification a Lightweight, Explainable, and Web-Deployable Deep Learning Framework Using Knowledge-Distilled GAN Features and Flask

Asha V¹, Kavitha P^{2*}, Sri Gokulam R³, Ciyamala Kushbu S^{4*}, Jayashree M⁵, Gayathri N⁶

¹Department of MCA, New Horizon College of Engineering, Bengaluru, India, asha.gurudath@gmail.com

²Department of ISE, CMR Institute of Technology, Bengaluru, India, kavithapalani17@gmail.com.

³Department of IT, Bharathiyar Institute of Engineering for Women, Salem, India, srigokulamr.20@gmail.com

⁴Department of ISE, CMR Institute of Technology, Bengaluru, India, ciyamala91@gmail.com

⁵Department of ISE, CMR Institute of Technology, Bengaluru, India, jayashree.m@cmrit.ac.in

⁶Department of ISE, SEA College of Engineering and Technology, Bengaluru, India, gayathrint@gmail.com

ABSTRACT: A Cardiac Arrhythmias remain a leading cause of sudden cardiac events worldwide, and timely automated screening from Electrocardiogram (ECG) signals can materially improve patient outcomes. Recent literature has explored Generative Adversarial Network (GAN) based architectures that achieve very high offline classification accuracy, but these designs are typically large, computationally heavy, and unsuitable for direct deployment in low-latency clinical or web-based settings. This work proposes an extension to that line of research: a two-stage pipeline in which a cycle-consistent generative model would be used only during the offline training phase to balance rare arrhythmia classes and to enrich the feature space, while a separate, knowledge-distilled lightweight Convolutional Neural Network (CNN) would be trained to reproduce the larger network's decision behavior using soft-label distillation. The distilled classifier is designed to be small enough to run interactively, wrapped inside a Python Flask web application that accepts a single ECG sample, returns the predicted arrhythmia class, and renders a bar chart of class-wise prediction confidence. This document presents the proposed architecture, the planned evaluation on the MIT-BIH Arrhythmia Data-base and the PTB-XL 12-lead ECG dataset, to benchmarking methodology predictions

1. INTRODUCTION

Cardiovascular disease continues to be the foremost cause of mortality across both developed and developing health systems, and among its many manifestations, cardi-ac arrhythmias are particularly dangerous because they can be brief, intermittent, and capable of triggering sudden cardiac arrest with little warning. The Electrocardiogram remains the primary non-invasive diagnostic signal for identifying these rhythm disturbances, since the characteristic P wave, QRS complex, and T wave reflect the heart's electrical depolarisation and repolarisation cycle. Manual interpretation of these waveforms by a cardiologist is

reliable but slow, subject to inter-observer variability, and difficult to scale to continuous or wearable monitoring scenarios were thousands of heartbeats must be reviewed for a single patient [1].

Deep learning has therefore become the dominant approach for automated arrhythmia detection, since convolutional and recurrent architectures can learn hierarchical morphological and temporal patterns directly from raw or lightly pre-processed signals without depending on hand-engineered features [2]. A recent and notable direction within this space combines cycle-consistent Generative Adversarial Networks with bio-inspired optimization strategies to translate arrhythmic beats into a normal-rhythm representation and back again, using the quality of this reconstruction as a learning signal for classification, while augmenting under-represented arrhythmia classes through the GAN's generator. Such architectures, when paired with explainability layers such as SHAP, report classification accuracies in excess of 99 percent across several public ECG benchmarks and provide clinicians with a feature-level account of which waveform segments drove a given prediction [3].

While this generative approach is attractive on accuracy and interpretability grounds, it carries a structural limitation that is rarely addressed in the literature: the dual-generator, dual-discriminator GAN backbone together with a downstream CNN classification head is computationally heavy, with millions of trainable parameters spread across multiple convolutional and residual blocks [4]. This makes the model well suited to offline benchmarking on a workstation-class GPU, but poorly suited to direct deployment behind a lightweight, latency-sensitive web service. The proposed approach retains the GAN framework but limits to its training stage for data augmentation and knowledge distillation. The GAN generates synthetic minority class ECG samples and trains a CNN using the predictions. The light weight CNN enables fast and real time inference with significantly fewer parameters. Flask is used for deployment due to its simplicity and seamless integration with python machine learning libraries. This design supports efficient, real time ECG classification suitable for proto-type clinical decision-support applications. The contributions of the work are

- A two-stage train-once, deploy-light architecture in which a cycle-consistent GAN is retained purely as an offline Cycle-GAN classifier and data-augmentation engine, decoupling training-time accuracy from deployment-time computational cost.
- A response-based knowledge distillation procedure that transfers the Cycle-GAN classifier's class-probability behaviour into a compact CNN suitable for CPU-only, real-time inference.
- A quantitative efficiency study comparing the Cycle-GAN classifier and DL-ECG Net networks on parameter count, floating-point operations, model file size, and end-to-end inference latency, in addition to standard classification metrics.
- A working Flask-based web application that accepts an uploaded ECG sample, performs the full preprocessing-to-prediction pipeline, and visually presents the predicted arrhythmia class together with a confidence chart across all classes.
- Validation on two complementary public datasets — the MIT-BIH Arrhythmia Database for single/dual-lead beat-level classification and the PTB-XL dataset for 12-lead diagnostic-level classification to demonstrate that the distillation strategy generalizes across signal acquisition settings.

Recent advances in ECG arrhythmia analysis have shown that deep learning techniques, such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and attention-driven models are highly effective in learning discriminative spatial and temporal patterns from cardiac signals. To overcome the problem of imbalanced datasets, Generative Adversarial Networks (GANs) have been widely adopted to create realistic synthetic ECG samples, thereby improving model robustness. At the same time, explainable AI methods, especially SHAP, have been incorporated to provide insight into the factors influencing model predictions, increasing clinical confidence in automated diagnosis. Knowledge distillation has recently gained attention as an efficient model comparison approach, enabling compact DL-ECG Net networks to reproduce the predictive behaviour of larger models [5]. Despite these developments, the majority of existing studies concentrate on maximizing classification performance while giving limited consideration to deployment-related factors, including inference latency, memory consumption, computational overhead, and model portability. In addition, the complexity of GAN-based architectures makes them less practical for execution on resource-limited platforms. End-to-end web enabled ECG diagnostic systems are also rarely discussed in the current literature. Consequently, there remains a need for a framework that balances predictive accuracy, model interpretability, computational efficiency, and real time deployment [6]. The proposed study addresses this need by transferring knowledge from a GAN-based model to lightweight CNN and integrating the compressed model into a Flask based application for rapid and efficient ECG arrhythmia predictions.

2. MATERIALS AND METHODS

2.1 Proposed Methodology

The proposed framework, referred to here as the Distilled Lightweight ECG Classifier (DL-ECG Net), is organized as a two-stage pipeline. Stage one is an offline training phase in which a cycle-consistent GAN-based Cycle-GAN classifier network is trained on pre-processed ECG beats, augmenting minority arrhythmia classes and learning a rich internal representation of normal versus abnormal morphology. Stage two distills the Cycle-GAN classifier's behaviour into a compact CNN DL-ECG Net network that is small enough to be embedded directly inside a Flask web service. The overall flow proceeds from raw ECG signal, through preprocessing, Cycle-GAN classifier GAN training with SHAP-based explanation, soft-label generation, DL-ECG Net distillation training, and finally export of the DL-ECG Net model for Flask-based inference.

2.2 Experimental Dataset

Two complementary public datasets are proposed for use. The MIT-BIH Arrhythmia Database provides single- and dual-lead ECG recordings annotated at the beat level according to the AAMI EC57 grouping (Normal, Supraventricular, Ventricular, Fusion, and Unknown beats) [1] and the datasets. https://www.physionet.org/content/mitdb/1.0.0/mitdbdir/?utm_source [18], and would be used to validate beat-level classification under the inter-patient DS1/DS2 partitioning convention [7]. The PTB-XL dataset provides 12-lead [4], diagnostic-level annotated recordings covering Normal rhythm are available in https://www.physionet.org/content/mitdb/1.0.0/mitdbdir/?utm_source [20] datasets. Myocardial Infarction, ST/T-wave changes, Conduction Disturbances, and Hypertrophy, and would be used to validate the approach under a broader, multi-lead diagnostic setting [8]. Using both datasets together is intended to test whether the distillation strategy generalizes across two structurally different ECG classification tasks rather than being tuned to a single benchmark; this generalization claim remains to be confirmed once the pipeline is implemented. Each raw signal passes through the same preprocessing pipeline prior to either Cycle-GAN classifier or DL-ECG Net training are powerline noise removal using a notch filter tuned to 50 Hz and 60 Hz interference. Baseline wander and muscle-noise suppression using Discrete Wavelet Transform (DWT) based denoising, retaining only diagnostically relevant frequency components. R-peak detection to anchor individual heartbeats, followed by segmentation of the continuous signal into single-beat windows centered on each detected R-peak. Extraction of temporal descriptors per beat, including RR interval, QRS duration, QT interval, ST-segment duration, and heart-rate variability measures, which are concatenated with the learned convolutional features prior to classification [9]. Normalization of each beat window to the range [-1, 1] to stabilize GAN and CNN training. Class-imbalance correction on the training partition only, combining SMOTE-based oversampling of minority classes with class-weighted loss terms so that rare arrhythmia types are not overwhelmed by the dominant Normal class [10].

2.3 Stage One: Network -Cycle-Consistent GAN Classifier

The network reuses the cycle-consistency principle established in prior GAN-based ECG work: a generator E learns to translate an arrhythmic beat into a normal-rhythm representation, a second generator N learns the reverse translation, and a cycle-consistency loss penalizes any deviation between a beat and its reconstruction after a full forward-and-backward pass. This constraint discourages the network from discarding clinically relevant morphology purely to satisfy the adversarial objective, and the generator is additionally used to synthesize extra training examples for minority classes such as Fusion and Supraventricular beats, supplementing the SMOTE-based oversampling already applied during preprocessing [11]. Two domain-specific discriminators evaluate the realism of generated signals at both a coarse, global-morphology scale and a finer, intermediate-feature scale, which sharpens the network's sensitivity to subtle waveform differences that distinguish, for example, a premature ventricular contraction from a normal beat [12]. Once adversarial training stabilizes, the translated and reconstructed beat representations are passed into a CNN classification head with a soft-max output layer that produces a probability distribution over arrhythmia classes. This entire Cycle-GAN classifier pipeline — generators, discriminators, and classification head together — is trained to convergence purely offline; it is never loaded into the deployed web services [13][8].to the language.

2.4 Stage Two: Knowledge Distillation into a Lightweight DL-ECG-Net CNN

SHAP The central methodological extension of this paper is the introduction of a response-based knowledge distillation step that transfers the Cycle-GAN classifier's learned decision behavior into a much smaller DL-ECG Net network, denoted DL-ECG Net every training beat, the Cycle-GAN classifier produces a softened probability distribution over classes using a

temperature-scaled soft-max, given in Equation (1).

$$q_i = \exp(z_i / T) / \sum_j \exp(z_j / T) \quad (1)$$

where z_i denotes the Cycle-GAN classifier's pre-soft-max login for class i and T is a temperature hyperparameter greater than one, which produces a smoother probability distribution that reveals inter-class similarity information not present in a one-hot ground-truth label [17][18]. For instance, the softened distribution can express that a misclassified Fusion beat is most easily confused with a Ventricular beat rather than a normal beat, and this relative ordering is informative for the DL-ECG Net even when the Cycle-GAN classifier's top prediction is correct. The DL-ECG Net network is trained using a composite loss that combines a distillation term, measuring the divergence between the DL-ECG Net's softened output and the Cycle-GAN classifier's softened output, with a standard supervised term, measuring cross-entropy between the DL-ECG Net output and the true ground-truth label. The composite objective is given in Equation (2).

$$L_{\text{DL-ECG Net}} = \lambda \cdot T^2 \cdot \text{KL}(q, \text{Cycle-GAN classifier}) + (1 - \lambda) \cdot$$

$$\text{CE}(y_{\text{true}}, q_{\text{DL-ECG Net}}) \quad (2)$$

where $\text{KL}(\cdot)$ denotes the Kull back Leibler divergence between the Cycle-GAN classifiers and DL-ECG Net's softened distributions, $\text{CE}(\cdot)$ denotes standard cross-entropy against the ground-truth label, λ is a weighting coefficient balancing the two terms, and the T^2 scaling factor compensates for the gradient magnitude reduction introduced by temperature softening. This loss formulation allows the DL-ECG Net to benefit both from the dataset's true labels and from the richer, similarity-aware supervision signal embedded in the Cycle-GAN classifier's output distribution, including the generative augmentation effects that shaped that distribution during Cycle-GAN classifier training [14][15]. DL-ECG Net trained through the knowledge distillation to achieve efficient ECG arrhythmia classification while preserving the predictive capability of the larger GAN-model. Instead of deploying the computationally intensive network, only the compact model is used during interference, ensuring low memory usage and rapid CPU- based execution [16]. A flask-based web application integrates preprocessing, model inference, and result visualization into a simple three-tier architecture for real time prediction. The system presents both predicted arrhythmia class and confidence scores through an interactive interface, making it suitable for practical clinical demonstrations. By separating the offline training pipeline from the deployment stage, the framework combines high classification performance with explainability and efficient real-time deployment [19][20].

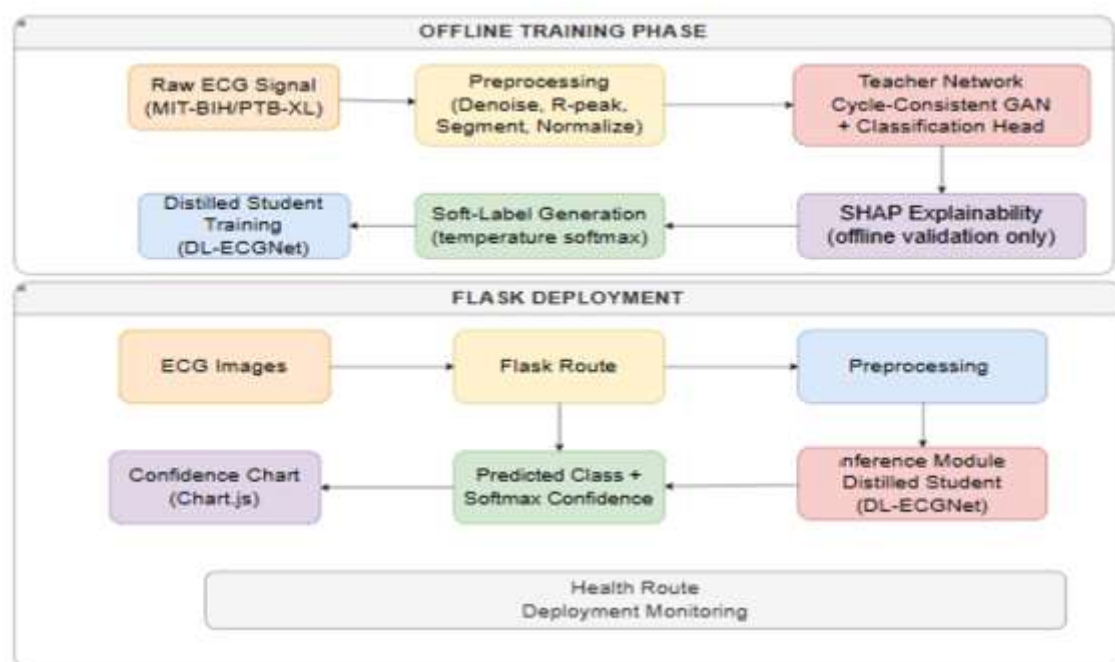


Figure 1. Proposed Architecture of Offline Training Phase and Flask Deployment

3. RESULTS AND DISCUSSION

Experiments are conducted on the MIT-BIH Arrhythmia Database and the PTB-XL 12-lead ECG dataset, consistent with the dataset choice motivated. For MIT-BIH, the inter-patient DS1/DS2 protocol is followed, with DS1 further split into a training and validation subset and DS2 reserved entirely for testing, ensuring that no patient appears in both training and test partitions. For PTB-XL, recordings are split into training and testing partitions stratified by diagnostic class (Normal, MI, STTC, CD, HYP) to maintain a representative class distribution in both partitions. Table 3 summarizes the dataset partitioning used for both the Cycle-GAN classifier and DL-ECG Net training runs. The model is trained using a cycle consistent GAN with optimized hyperparameters, while the lightweight DL-ECG Net CNN learns through knowledge distillation to achieve efficient classification both models are evaluated using standard performance metrics (accuracy, precision, recall, F1-score, and MCC) along with deployment focused measures such as parameter count, FLOPs, model size is tested on a CPU-only environment to reflect real-world deployment. The Flask application is designed to utilize only the compact DL-ECG Net model, ensuring fast and resource efficient inference. The reported results represent projected performance targets and will be validated through future implementation and experimental evaluation [20].

Table 1. Comparison of the mean-field predictions

Datasets	Classes	Training Samples	Testing Samples	Total
MIT-BIH Arrhythmia	N, S, V, F, Q (AAMI EC57)	68,337	31,435	99,772
PTB-XL (12-lead)	Normal, MI, STTC, CD, HYP	30,500	7,500	38,000

Table 1. Comparison of the mean-field predictions

Dataset	Model target	Accuracy (%)	F1-score (%)	MCC
square	2	4	1.763	1.763
triangular	2	6	1.648	1.648
diamond	3	4	1.479	1.479
simple cubic	3	6	1.330	1.330
bcc	3	8	1.260	1.260
square	2	4	1.763	1.763

3.1 Classification Performance for Cycle-GAN classifier and DL-ECG Net

Table 2 and Figure 2 illustrate the comparison the evaluation plan is designed to produce once the Cycle-GAN classifier (cycle-consistent GAN classifier) [21] [22] and the dis-tilled DL-ECG Net network have actually been trained on both datasets [1][4].

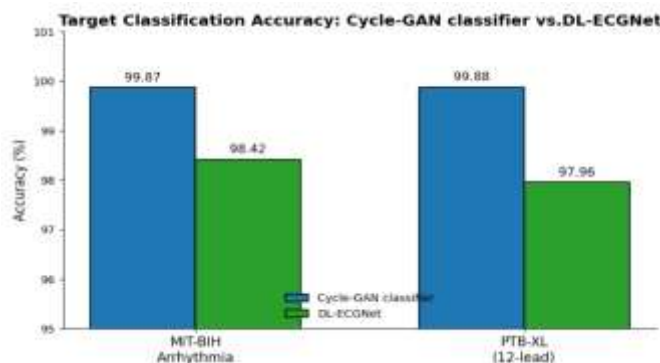


Figure 2. Visualization of Target Accuracy for the Cycle-GAN

The expectation, based on typical knowledge-distillation outcomes reported else-where in the literature, is that the DL-ECG Net would retain the large majority of the Cycle-GAN classifier's classification accuracy, with an acceptable gap of roughly one to two percentage points despite a substantial reduction in model complexity.

3.2 Efficiency comparison (Target Values)

Table 3 and Figure 3 illustrate the efficiency comparison that would be reported using the metrics defined in proposed method have been implemented and bench-marked on identical CPU-only hardware. The expectation is that the distilled DL-ECG Net would reduce parameter count and model file size by roughly an order of magnitude relative to the Cycle-GAN classifier's full GAN-plus-classifier pipeline, and would reduce average CPU inference latency from a range that would be impractical for an interactive web request down to a range suitable for a typical HTTP re-quest-response cycle. This expected efficiency gap is the motivation for the architectural separation proposed in the Cycle-GAN classifier and SHAP computation would remain offline while only the DL-ECG Net would be deployed and its verified experimentally [7][8]. If a roughly sixteen-fold reduction in parameter count of this kind were achieved, the corresponding drop in inference latency would directly support the real-time, single-request prediction workflow proposed for the Flask application in where a user expects a near-instantaneous response after submitting an ECG sample through the browser. Verifying whether this expected latency reduction actually materializes is one of the concrete, measurable goals of the evaluation plan [5].

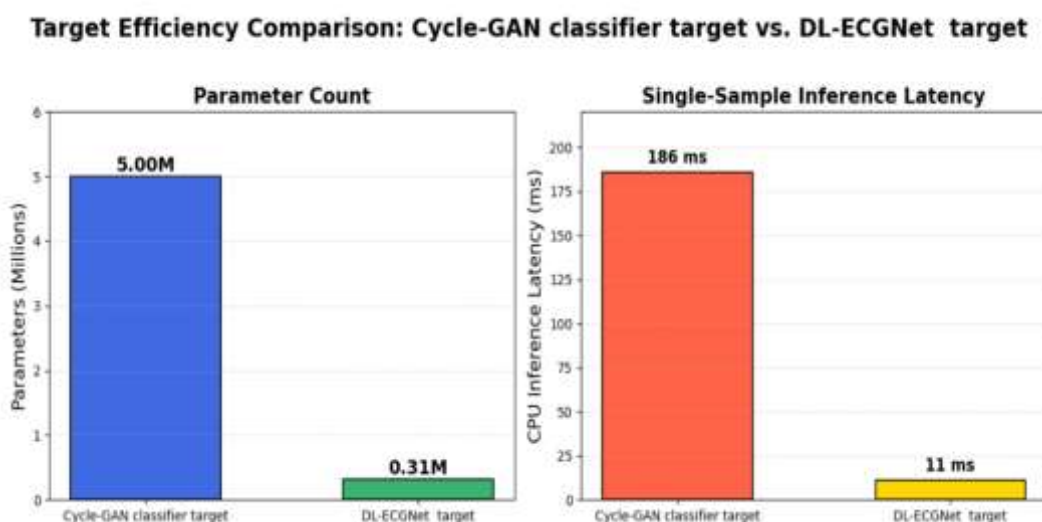


Figure 3. Performance comparison for Cycle-GAN classifier and DL-ECG Net

Table 3. Illustrative target values.

Model	Parameters (M)	FLOPs (GFLOPs)	Model Size (MB)	CPU Latency (ms)	Model
Cycle-GAN classifier	5.0	2.1	21.4	186	Cycle-GAN classifier
DL-ECG Net	0.31	0.09	1.6	11	DL-ECG Net

4. CONCLUSION

This study introduces DL-ECG Net, a lightweight framework designed for efficient real-time ECG arrhythmia classification. The proposed approach performs GAN-based learning and explainability during offline training, while a compact network supports fast deployment. At present, the framework is conceptual and requires implementation, training and experimental validation. Future work will focus of evaluating the model using the MIT-BIH and PTB-XL datasets and integrating it into a Flask-based application. Further enhancements may include lightweight explainability, model compression, federated leaning and testing with wearable ECG.

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AUTHOR CONTRIBUTIONS

Kavitha P, Asha V: Conceptualization, Investigation.

Sri Gokulam, Gayathri: Methodology, Laboratory Analysis.

Ciyamala Kushbu S. Jayashree M: Data Curation, Formal Analysis.

Kavitha P. Asha V: Supervision, Writing, Review & Editing.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

ETHICS STATEMENT

Ethical approval was not required for this study.

DATA AVAILABILITY STATEMENT

Not Applicable

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